YUVANESH KS

Software Engineer • Cloud & Linux Infrastructure • Cross-Platform Dev

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PROFESSIONAL SUMMARY

Hands-on **Software Engineer** specializing in Linux systems, self-hosted virtualization infrastructure, automated CI/CD delivery pipelines, and reactive cross-platform Flutter application architecture. Proven engineering record deploying production commercial server infrastructure (Bunny.net CDN, Docker Compose), published Google Play Store applications (100% production rollout), and open-source quantum machine learning research.

TECHNICAL COMPETENCIES

Core Languages:	**Java (DSA)**, **Python**, **Dart**, Bash Scripting, C/C++, SQL, HTML5/CSS3
Systems & DevOps:	**Linux Administration** (Arch, Rocky, Debian/Ubuntu), **Docker Compose**, **KVM/QEMU**, Git CI/CD, Bunny.net CDN, Tailscale, Hetzner VPS
Applications & Design:	**Flutter**, WebRTC P2P Data Channels, mDNS LAN Discovery, AES-GCM Payload Encryption, **Vanilla CSS Design Tokens (ui-themes)**, PocketBase
AI & Research:	**PyTorch (GRU/CNN)**, **ONNX Runtime**, **Streamlit**, PennyLane (Quantum ML / Qutrits), XGBoost, NASA C-MAPSS RUL, SHAP Explainability

ENGINEERING EXPERIENCE

Starvik Solution Pvt Ltd — Flutter Internship Team Lead & Infrastructure Engineer — **Jun 2026 – Jul 2026**

Company Site ↗ | GitHub Profile ↗ | Flutter | WebRTC | Bunny.net CDN | Docker | Git CI/CD

- Architected and delivered **Abyss Chat**, a decentralized cross-platform P2P communications suite featuring WebRTC audio/video call orchestration, mDNS local discovery, and AES-GCM encryption.
- Engineered corporate server infrastructure, continuous deployment (Git CI/CD), edge storage routing with Bunny.net CDN, and environment secret shielding for Starvik's web ecosystem.
- Mentored engineering interns on clean reactive state management, asynchronous Dart concurrency, and production build pipelines.

Indian Data Club (IDC) — Joint Secretary • AMCET Chapter — **Aug 2025 – Present**

Community ↗ | Technical Leadership | Data Science | Hackathons | Workshops

- Co-directing technical initiatives and workshops for a pan-India student data science community, organizing code sprints, hackathons, and AI peer-learning sessions.

1M1B • AICTE • Salesforce — Green & Tech Intern (Batch 5) — **Aug 2025 – Oct 2025**

Credentials ↗ | Salesforce | Green Tech | Impact Analytics

- Selected among top applicants across India for sustainability-driven tech exploration, researching digital solutions for environmental impact tracking.

Virtualization & Enterprise Home Lab — Systems Architecture Project — **2025 – 2026**

Lab Configs ↗ | KVM/QEMU | Rocky Linux | Arch Linux | Docker | Bash Automation

- Configured multi-OS virtualized test environment on Rocky Linux and Arch using KVM/QEMU, containerizing localized services with Docker and automating provisioning with Bash.

RELEASED PRODUCTS & FEATURED PROJECTS

Abyss Chat — Decentralized P2P Mesh Communication Suite — **2026**

Google Play 100% Rollout | GitHub ↗ | Live Web ↗ | Play Store ↗ | Flutter | WebRTC | mDNS LAN | AES-GCM

- Engineered a server-independent P2P messaging and video application utilizing Flutter, WebRTC peer data channels, PeerJS STUN servers, and mDNS LAN node discovery.
- Achieved 100% production rollout on the **Google Play Store** after clearing Google's strict developer privacy and technical requirements.

SkyPrognos — Aviation Fleet Health & Predictive Maintenance — Predictive AI Platform — **2026**

Live Digital Twin | GitHub ↗ | Live Demo ↗ | Python | PyTorch (GRU/CNN) | ONNX Runtime | Streamlit | Docker

- Forecasts turbofan Remaining Useful Life (RUL) with ~11.9 RMSE on NASA C-MAPSS telemetry; exported PyTorch models to ONNX for ~2.7ms CPU inference.
- Architected interactive Streamlit command center with fleet ranking, live replay, and an idempotent SQLite outbox simulating edge telemetry sync.

Universal UI Themes & Design Tokens — Zero-Dependency Design System — **2026**

Open-Source Library | GitHub ↗ | Live Demo ↗ | Vanilla CSS | Design Tokens | Claymorphism | Glassmorphism

- Authored 22 production-ready theme presets (Material 3, Claymorphism, Tokyo Night, Catppuccin) in pure Vanilla CSS with 3-tier token architecture.
- Enables 1-line dynamic theme switching via HTML attribute with zero runtime dependencies, portable across Web and Mobile UI tokens.

Q-Ternary VQC: Logarithmic Qudit Compression — Quantum Machine Learning — **2026**

Open-Source Research | GitHub ↗ | Publication ↗ | PennyLane | PyTorch | 3-Level Qutrits | SHAP Biomarkers

- Formulated hybrid Quantum-Classical Variational Quantum Classifier (VQC) in PennyLane/PyTorch utilizing 3-level Qutrits, reducing feature footprint by 33% without information loss.
- Achieved 88.60% clinical accuracy with 72 trainable parameters, integrating SHAP biomarker attribution for explainable quantum utility on standard CPU hardware.

Trackify — Cross-Platform Habit Tracking Product — Productivity Application — **2026**

Released Across 4 OS | GitHub ↗ | Live Web ↗ | Flutter | Android | Linux | Windows | GitHub Actions CI/CD

- Shipped high-performance privacy-first tracking application deployed to Web, Android, Linux, and Windows desktop with Provider and SharedPreferences.

EDUCATION

Annai Mira College of Eng. & Tech. — **2023 – 2027**

B.Tech in Information Technology (Anna University)

CGPA: 8.26 • 2× Smart India Hackathon (SIH) Participant • Databricks AI Challenge • NASA Space Apps • GFG Vultr Hackathon

CERTIFICATIONS & HONORS

- **Oracle:** Cloud Infrastructure Generative AI Professional • [Verify ↗]
- **Google:** Cybersecurity Professional Certificate • [Verify ↗]
- **Cisco:** CyberOps Associate • Networking Essentials • [Verify ↗]
- **AWS:** Cloud Quest: Cloud Practitioner • [Verify ↗]